Lecture 29

RNA-Seq Data Analysis

RNA-Seq is a quantification

Most quantifications follow the workflow

1. Align
2. Quantify
3. Compare

There are many variations for each step.

RNA-Seq quantifies transcripts

Transcriptions is ... complicated

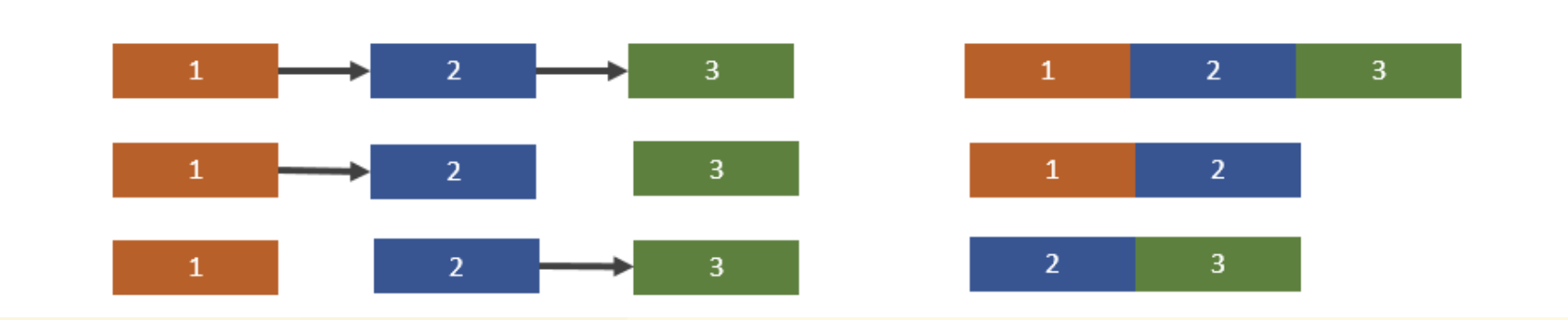
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1. Bacteria: operons, termination regulates the transcription
2. Eukaryotic cells: alternative splicing

Junction: a read that bridges over an apparent "gap" in the genome a fusion of two exons.

Isoforms

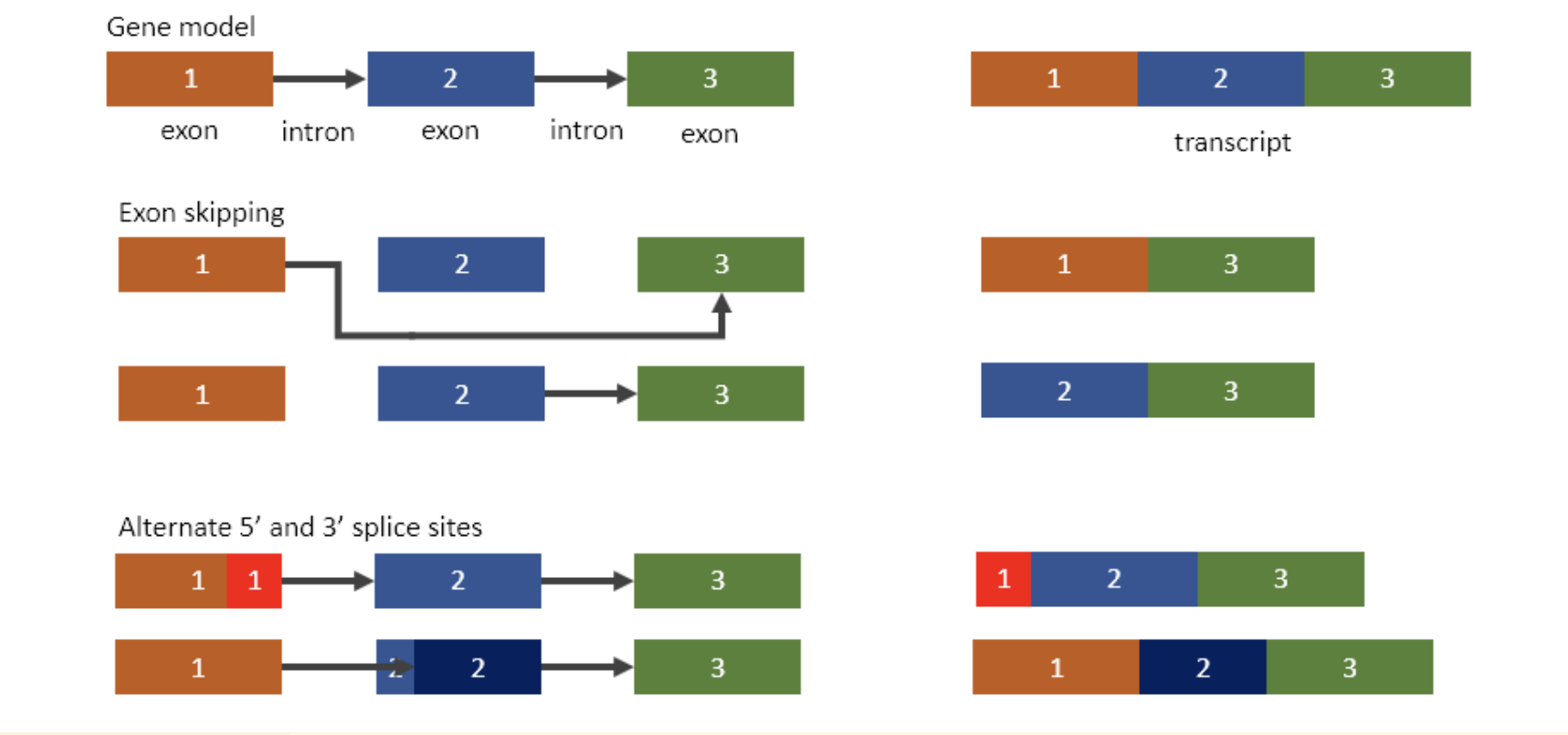
The same pieces of DNA may be assembled into different linear transcripts



You can detect transcripts by the reads that span over a change `1->2`, `2->3` etc.

Alternative splicing

The more common splicing mechanisms



RNA-Seq nomenclature

An RNA-Seq analysis has three separate yet equally important segments:

1. Identifying transcripts
2. Estimating abundances per transcript
3. Comparing abundances --> differential expression

There is disagreement about how each these steps should be performed -> hence a large number of options.

There are workflows that mix and match from differnt methods.

1. Alignment Step

First choice in RNA-Seq analysis

Red pill or blue pill:

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1. Quantify against a genome

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2. Quantify against a transcriptome

Think about advntages and disadvantages of each.

Then you have a variety of choices for each.

You chose to use a genome as reference

Pros:

1. It does not need an annotation (though annotations help)
2. It can discover novel transcripts

Cons:

1. Less accurate. It is more difficult to resolve ambigously alignments
2. Non-expressed regions may influence the alignment

You chose a transcript as reference

Pros:

1. Does not require a fully assembled accurate genome
2. Better quantification for similar transrcipt

Cons:

1. Can't find novel transcripts
2. Requires good quality transcript information.

Gene level analysis

What takes place when someone performs a **gene** level analysis?

We estimate abundance over the sum of all exons that exist.

Make one long transcript built from all exons and call that the **gene**.

Caveat: This "theoretical" transcript does not have to exist in this form.

Are gene level analyses valid?

In some cases yes.

When the phenotype is such that relative abundance of alternative trascripts don't matter.

When the phenotype is dominated by transcripts that belong to from different genes.

Data Replication

mRNA abundances is dynamic, it changes over time within a cell.

We need to ensure that any change we measure is due to the condition change and not the normal variability.

We need to make multiple measurements for the same condition.

To detect a change the variation across replicates has to be smaller than the variation between conditions.

Higher coverage or more replicates?

An evolving concept with tradeoffs.

The more data we collect the more accurate an individual estimate.

The more replicates we have the better we asses the natural variability.

Current recommendation:

More replicates are better: 4 or more.

2. Abundance estimation

How should we count an ambigous alignment?

This problem occurs only when aligning against a genome.

You may reads that

1. Are not fully contained in a transcript
2. May align in multiple locations
3. Appear to be of low quality

Different counting (abundance estimation) strategies may produce different results.

It is worth testing other strategies

Counting is usually not that time consuming compared to other steps.

It is easier to re-run another strategy if the process is automated.

3. Abundance comparison

Comparing abundacnes

Normalization: the process of making data comparable.

The most common questions to be answered:

Is `Gene A` expressed in more copies than `Gene B` within `condition 1`?

Is `Gene A` expressed in more copies in `condition 1` vs `condition 2`?

A tortous history of abundance estimation

Suprisingly convoluted history (see Book for details):

1. Counts per million
2. RPKM: Reads per kilobase per million mapped reads
3. FPKM: like RPKM but for fragments
4. TPM: Transcripts per million
5. TMM: A statistical concept that estimates scale factors.

Each was designed to "protect" biologists from mathematics - only to end up being more difficult to

What should I use?

1. RPKM and TPM are ok to use to compare transcripts within a sample
2. Comparing across sample: TMM normalized counts

In class demo

We'll now follow the chapter

- **RNA-Seq: Griffith Test Data**

You will need R to be installed with proper packages.

See the chapter on **RNA-Seq Statistics.**

Megatons all the way.

A fully automated RNA-Seq data analysis pipeline:

```
# Get the data.
bash griffith-getdata.sh

# Align the data.
bash griffith-align.sh

# Estimate abundances.
bash griffith-count.sh
```